Quadrupole mass filtering in MScan

Rod voltages, scan law and parameters

ESIBD Explorer Plugins, v0.5.9

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1 Scope

MScan sweeps the RF amplitude A of one AMX output (CH0-CH1 or CH2-CH3) and, optionally, the DC of the two rod pairs through two AMPR channels. At each amplitude it averages the current of every enabled DMMR module, the ion collector among them, and saves current versus A for each. It does not change the AMX frequency, duty cycle or routing, and it does not convert A to m/z: the file states `calibration: None; amplitude in V, not m/z`.

2 Rod potentials

The PSU rails of the selected AMX output are set to $V_{\mathrm{pos}} = +A$ and $V_{\mathrm{neg}} = -A$; the AMX switches each rod pair between them in opposite phase, so with $s(t) = \pm 1$ a square wave of 50 % duty at the AMX frequency $f = \Omega/2\pi$:

$$\Phi_1(t) = x + U + A\,s(t) \qquad \text{pair 1 (AMPR channel Rod pair 1)} \tag{1}$$

$$\Phi_2(t) = x - U - A\,s(t) \qquad \text{pair 2 (AMPR channel Rod pair 2)} \tag{2}$$

These are equations (1) and (2) of Steel and Henchman [1] ($V_{2,4} = +U + V_0 \cos \omega t$, $V_{1,3} = -U - V_0 \cos \omega t$) with $V_0 = A$, the cosine replaced by $s(t)$, and a common potential x (the axis). Adding x to all four rods does not change the filter: it sets the potential of the axis, hence the axial energy of the ions. Which pair receives $+U$ does not matter either: with 50 % duty, swapping the pairs only shifts the wave by half a period. Figure 1 shows an example.

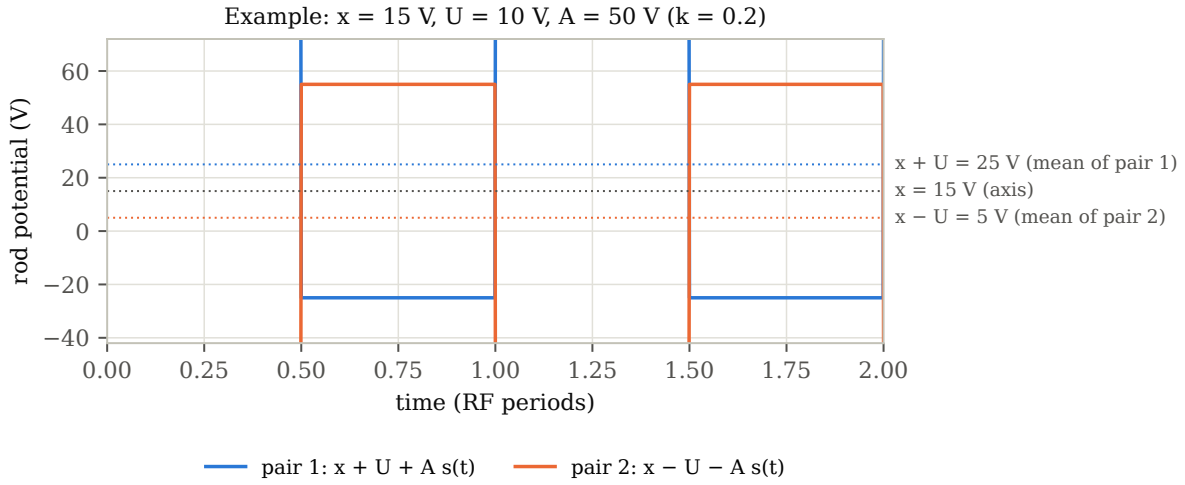


Figure 1: Rod potentials at one scan point. Only the antisymmetric part ($\pm U$, $\pm A\,s$) filters masses; x is common to both pairs.

Assumptions not yet verified on the instrument. (i) Each rod swings between $+A$ and $-A$ around its DC, i.e. $V_0 = A$ (the AMX switches between the PSU rails, referenced to the DC applied to that pair). (ii) The AMPR DC reaches the rods of the pair driven by the same AMX connector. (iii) The wave on the rods stays square. A scope on the rods (or on the AMX *Monitor* outputs, 1:100, if the AMX HD has them as the 4ED does) settles all three. The lab scan of 2026-11-02 14:34 supports (ii): the collector current responds to the difference between the two DC values and peaks when they are equal.

3 DC law and parameters

For a coefficient $k > 0$, MScan commands at every amplitude

$$U(A) = \max\bigl(U_0,\ k\,A - \delta\bigr), \qquad \Phi_{1,2}^{\mathrm{DC}} = x \pm U(A). \tag{3}$$

$k = 0$ is RF-only: both pairs stay at x, without the floor U_0. A negative k swaps the pairs (U is computed with $|k|$). With $\delta = 0$ and $k\,A \geq U_0$, equation (3) is the scan line of the textbook [1]: $U/A = k$ constant, a straight line through the origin.

Symbol	MScan setting	Unit	Default	Role
A	Amplitude from / to / step	V	–	scanned RF amplitude, per rod (rails $\pm A$)
x	Axis potential x	V	0	common potential of the four rods, i.e. the axis; sets the ion energy, does not filter. Keep the value the rods hold outside a scan.
k	U/V ratio k	–	0.2	slope U/A of the scan line; sets the resolution. 0 = RF-only, negative swaps the pairs.
δ	Resolution δ	V	0	lowers the line: $U/A = k - \delta/A$ rises with A, so peaks keep about the same width in A (Section 4). Must be ≥ 0.
U_0	Minimum U	V	0.5	floor of U, must be > 0 (Section 4).
–	Rod pair 1 / Rod pair 2	–	None	AMPR channels of the two pairs. Pair 2 alone is refused.

Table 1: Parameters of the DC law, in the Mass filter group of the MScan panel, which also shows a Preview of U and of both pair potentials over the scan. Settings files and HDF5 scans keep the earlier keys: Fixed offset (x), Offset coefficient (k), Resolution offset (δ), Quadrupole offset and Quadrupole offset pair 2.

Do not confuse k with the α of Steel and Henchman. Their α is the Mathieu parameter $a = 8eU/(mr_0^2\Omega^2)$; the ratio they keep constant is $U/V_0 = \alpha/(2q) = 0.165$ for a cosine. Here $k = a/(2q)$, and its useful values are near 0.2 because the wave is square (Section 4). Entering 0.233 as k would put the line above the apex: nothing passes.

One pair only. If only pair 1 is selected, pair 2 keeps its own value V_2. The filtering DC is then $(x + U - V_2)/2$ and the axis moves to $(x + U + V_2)/2$ during the scan. Use both pairs for equation (3).

At each point. The rails are commanded first, then both rod pairs (rounded to the AMPR display precision). No current is acquired until the AMPR confirms each target by its setpoint readback and its Monitor is within Voltage tolerance. Start is refused if a value is not finite, $\delta < 0$, $U_0 \leq 0$, a target over the span leaves the channel Min/Max or AMPR rating, or a channel is OFF or under equation control. Normal completion restores the initial pair potentials with the PSU setpoints; Stop or an error holds the last values.

Recorded. The setup keeps `offset = {coefficient, fixed_v, resolution_offset_v, minimum_u_v, pairs}` (channel, AMPR module and channel, sign, initial value)}; every point keeps the commanded pair potentials `offset_target` and their Monitors `offset_v`, one column per pair, next to `rail_v`, `rail_i`, sample counts and point status.

4 What the parameters do

With the conventions of [1] (U and $V_0 = A$ per rod, from the axis),

$$a = \frac{8eU}{mr_0^2\Omega^2}, \qquad q = \frac{4eA}{mr_0^2\Omega^2}, \qquad \frac{a}{q} = 2\,\frac{U}{A} = 2k \quad (\delta = 0). \tag{4}$$

All ions lie on the line $a = 2kq$; an ion passes when its point is inside the stability region (Figure 2). For a 50 % square wave the apex is at $a = 0.2381$, $q = 0.5529$, i.e. $k = 0.2153$ (cosine: 0.2370, 0.7060, $k = 0.1678$). The RF-only limit is $q = 0.712$ (cosine 0.908).

k	transmitted q band	centre $\bar{q}$	$R = \bar{q}/\Delta q$
0	0 – 0.712	–	high-pass
0.10	0.246 – 0.629	0.438	1.1
0.15	0.375 – 0.593	0.484	2.2
0.18	0.455 – 0.574	0.515	4.3
0.19	0.482 – 0.568	0.525	6.1
0.20	0.510 – 0.562	0.536	10.3
0.205	0.524 – 0.559	0.542	15.4
0.21	0.538 – 0.556	0.547	30.4
0.2125	0.545 – 0.554	0.550	57

Table 2: Transmitted q band versus k ($\delta = 0$, ideal 50 % square wave). Near the apex $R \approx 0.157/(0.2153 - k)$. R is $m/\Delta m$ of the band at fixed A.

Peak position. An ion transmitted at the centre $\bar{q}$ of the band has

$$\frac{m}{z}\,[\mathrm{Th}] \approx 2.444\,\frac{A\,[\mathrm{V}]}{\bar{q}}\left(\frac{4\,\mathrm{mm}}{r_0}\right)^2\left(\frac{500\,\mathrm{kHz}}{f}\right)^2, \tag{5}$$

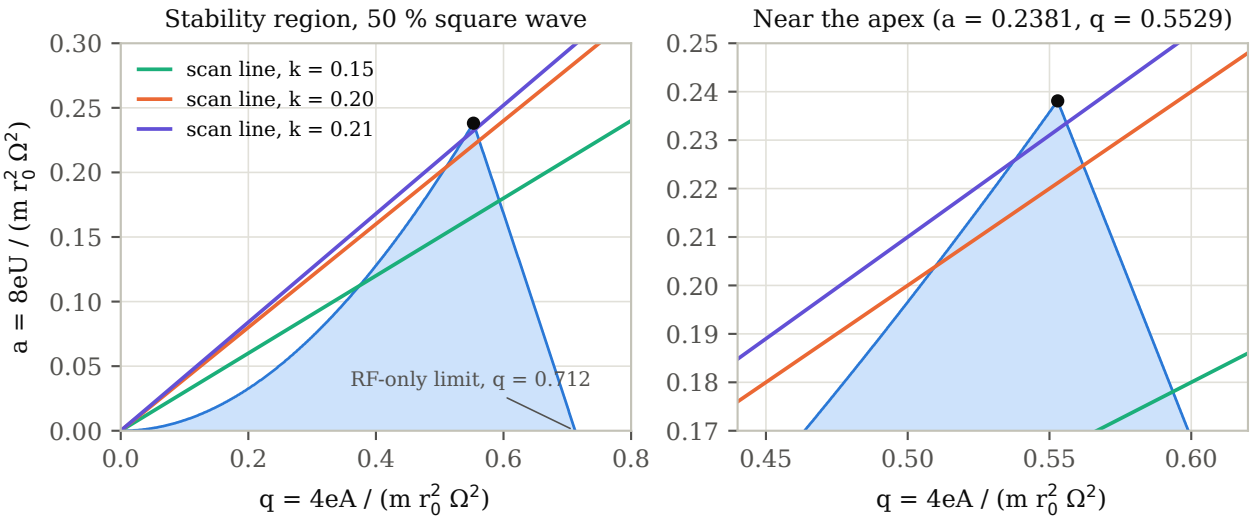


Figure 2: Stability region of the 50 % square wave (both directions stable) and scan lines $a = 2kq$. Computed with the transfer matrix of the piecewise-constant Hill equation.

about 4.56 A at $k = 0.2$ for $r_0 = 4$ mm and 500 kHz. MScan computes it at every point, with $\bar{q}$ from U/A there, the AMX frequency and the r0 setting (5 mm for Q2, provisional), shows it as m/z range and on the plot (Plot axis), and saves it as `Validation/mz`. Calibrating r_0 on one known ion also absorbs the unverified ratio V_0/A. In RF-only mode ($k = 0$) ions with $m/z < 3.43\,A$ (same units) are rejected, so a spectrum of several species is a staircase: each species removes a step at its cut-off amplitude A_c, and its peak at $k = 0.2$ lies near $0.753\,A_c$. The mass window scales as A/f^2: at fixed $A_{\max}$, 300 kHz reaches 2.8 times and 200 kHz 6.3 times the m/z of 500 kHz. The rail current also scales with f (Section 5).

Resolution and the amplitude on the rods. Table 2 assumes $V_0 = A$. If the rods see cA, the filter works at k/c: at $k = 0.20$ it closes completely for $c < 0.929$, at $k = 0.21$ for $c < 0.975$. The scan itself measures c: raise k until the signal disappears at k_c; then $c = k_c/0.2153$. Rounded edges matter much less: a trapezoid with rise times of 100, 200, 300 and 500 ns at 500 kHz moves the apex to $k = 0.2141$, 0.2111, 0.2064 and 0.1921; an exponential edge with $\tau = 50$, 100, 150 ns to 0.2123, 0.2044, 0.1937. With DC coupling, a duty error of 1 % moves the apex by about ±0.018 in k.

Resolution δ. With $\delta = 0$ the peak width grows in proportion to A (constant R). With k at the apex value and $U = kA - \delta$, the width is nearly constant, $\Delta A \approx 6.4\,\delta$ (simulated: 6.8, 6.5, 6.4, 6.4 V at $A = 50$, 100, 200, 300 V for $\delta = 1$ V). Commercial instruments set slope and intercept on a light and a heavy reference ion, because the ideal constant-width line is a curve [2]. A negative δ (a positive constant added to kA) at the apex slope passes nothing, which is why MScan refuses $\delta < 0$.

Minimum U. At zero DC every ion heavier than the RF-only cut-off passes. Without a floor, a scan starting near $A = 0$ begins with an unfiltered plateau (simulated for an ion with its peak at 300 V and $\delta = 1$ V: transmitted from 0 to 4.72 V with $\max(0, \cdot)$, only from 4.57 to 4.72 V without clamping). Commercial residual gas analysers keep the DC from reaching zero ("Zero Blast Suppression" [3]). Peaks below $A^* = (U_0 + \delta)/k$ are not filtered reliably.

5 Limits of the present set-up

- Rail current of the Q2 output (AMX HD, 500 kHz, lab 2026-11-02): about 12 mA + 0.565 mA/V × A per rail, i.e. 153 mA at 250 V. With $I_{\text{lim}} = 200$ mA the amplitude cannot exceed about 333 V at 500 kHz, even with the PSU range switched on. The current scales with f: about 97 mA at 300 kHz and 250 V.
- PSU headroom (PSU_A in Half range, lab 2026-11-03): output + dropout fell from 256 V at 0 mA to about 248.5 V at 150 mA, so the rails could not regulate above about 248 V and a 50–300 V scan stopped at 250 V. MScan now refuses a scan above output + dropout − 2 V, stops at once when a rail has lost its headroom, and flags points measured with less than 2 V (`low_headroom`). Full range raises the internal supply; check the dropout and the PSU temperature at the highest amplitude.
- The scan law is applied by two instruments (PSU and AMPR) that must agree to better than 1 % in ratio near the apex. Rectangular-wave (digital) quadrupoles can instead keep the amplitude and the DC fixed and sweep the frequency: U/A is then constant by construction and $m/z \propto 1/f^2$. MScan does not do this.
- A filter without DC is also possible with DC-coupled square waves by changing the duty cycle (computed: $R \approx 7$ at 60.0 %, 12 at 60.5 %, 41 at 61.0 %; closed above about 61.2 %). MScan does not do this either; it leaves the AMX timing unchanged.

6 Recommended order on a new set-up

1. Confirm ions on the collector at a fixed amplitude: the current must be positive (for positive ions) and fluctuate with the spray. The electrospray current alone does not prove that ions enter.
2. Record one scan with the electrospray HV off: it gives the background of the collector, which MScan does not subtract.
3. Scan with $k = 0$ to the highest available A: count the cut-off steps and note their A_c.
4. Scan with $k = 0.15, 0.18, 0.20, 0.21$ ($\delta = 0$, $U_0 = 0.5\,\mathrm{V}$); the value where the signal vanishes gives c (amplitude on the rods / A).
5. Only then use δ, set on two reference ions, if constant peak width is needed.

References

[1] C. Steel, M. Henchman, *Understanding the Quadrupole Mass Filter through Computer Simulation*, J. Chem. Educ. 75 (1998) 1049.

[2] R. E. Pedder, *Practical Quadrupole Theory: Graphical Theory*, Extrel Application Note RA_2010A (ASMS 2001).

[3] Stanford Research Systems, *RGA120/220/320 Residual Gas Analyzer, User Manual*, rev. 1.08 (2025), section "Zero Blast Suppression".